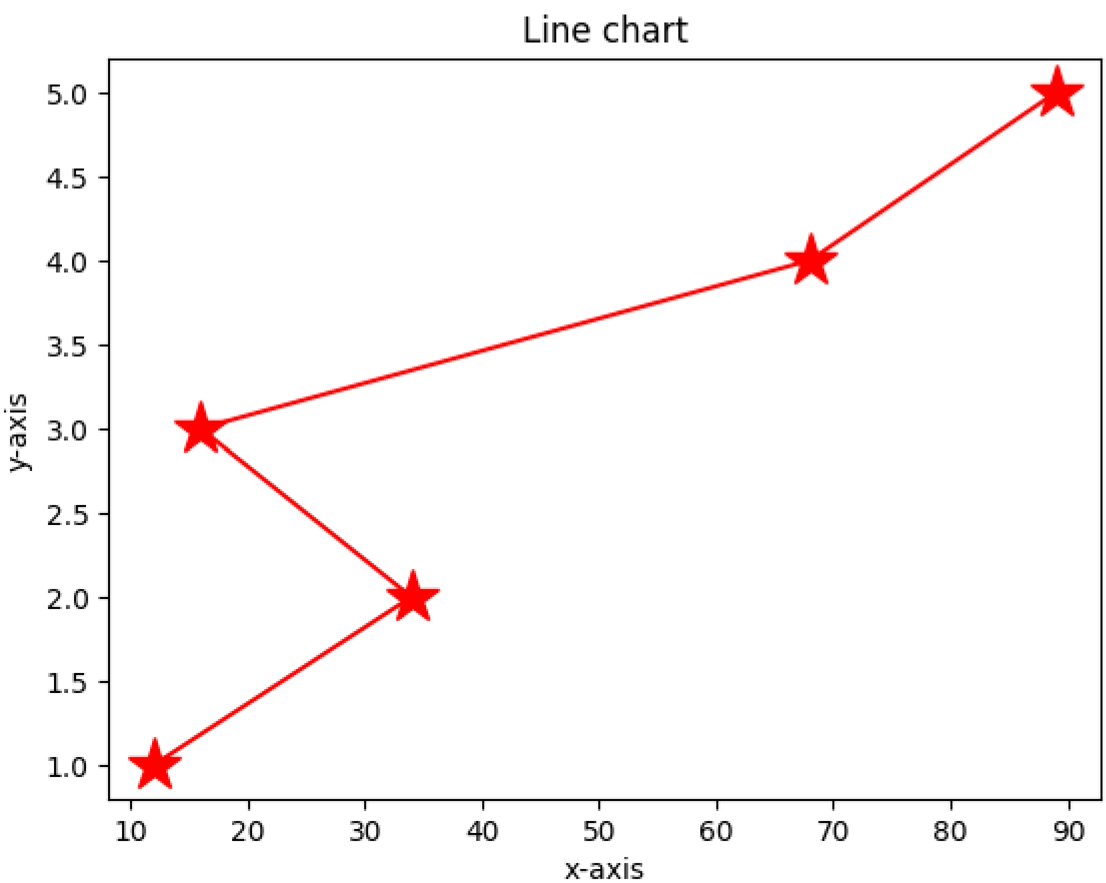
In [12]:
```
# pip install matplotlib
```

In [13]:
```
import matplotlib.pyplot as plt
```

In [16]:
```
x=[12,34,16,68,89]
y=[1,2,3,4,5]
```

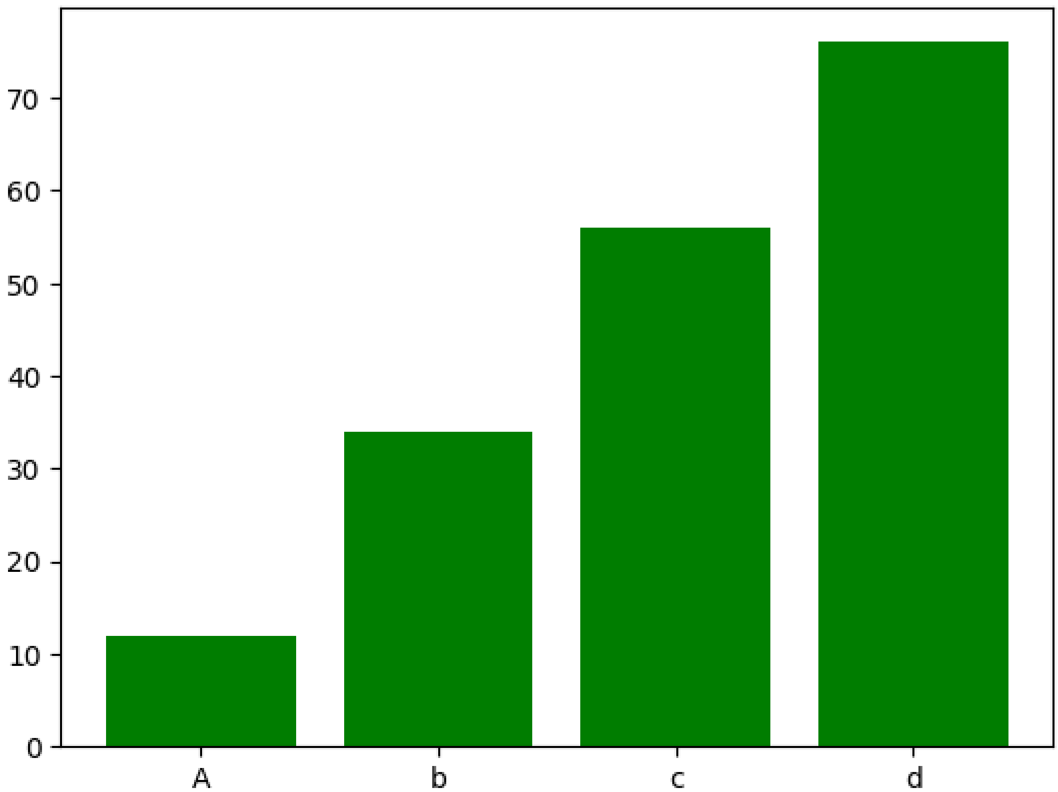
In [32]:
```
plt.plot(x,y, color= "red", linestyle= "solid", marker="*", markersize=20) #'-
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("Line chart")
```

Out[32]:
```
Text(0.5, 1.0, 'Line chart')
```

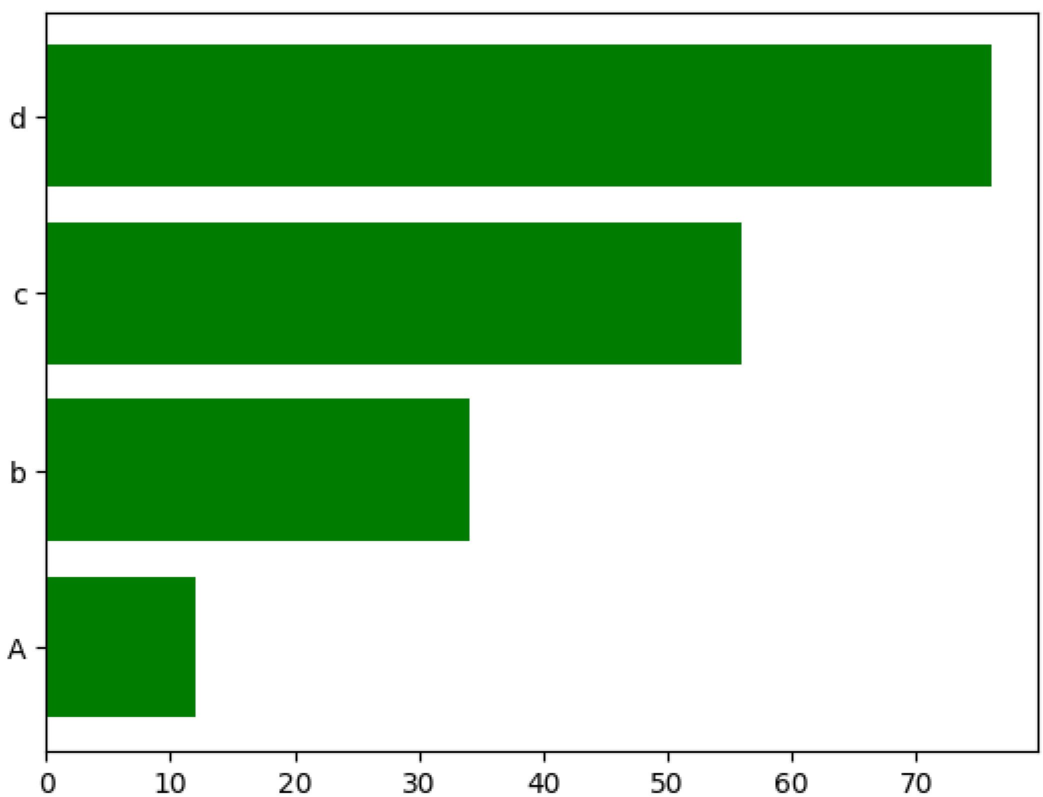


In [33]:
```
x2=["A","b", "c", "d"]
y2=[12,34,56,76]
```

In [36]:
```
plt.bar(x2,y2, color= "green")
plt.show()
```

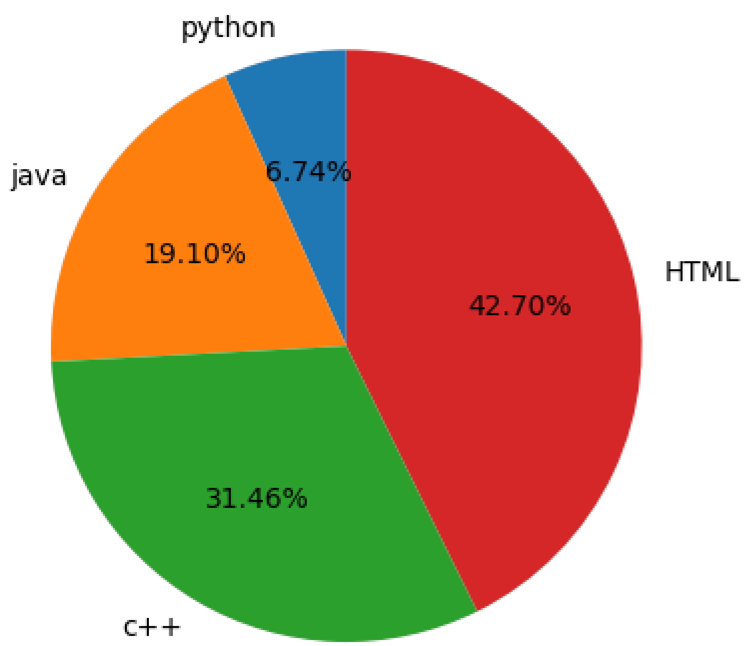


In [37]:
```
plt.barh(x2,y2, color= "green")
plt.show()
```



In [38]:
```
x2=["python","java", "c++", "HTML"]
y2=[12,34,56,76]
```

In [44]:
```
plt.pie(y2,labels=x2, autopct="%1.2f%%", startangle=90)
plt.show()
```



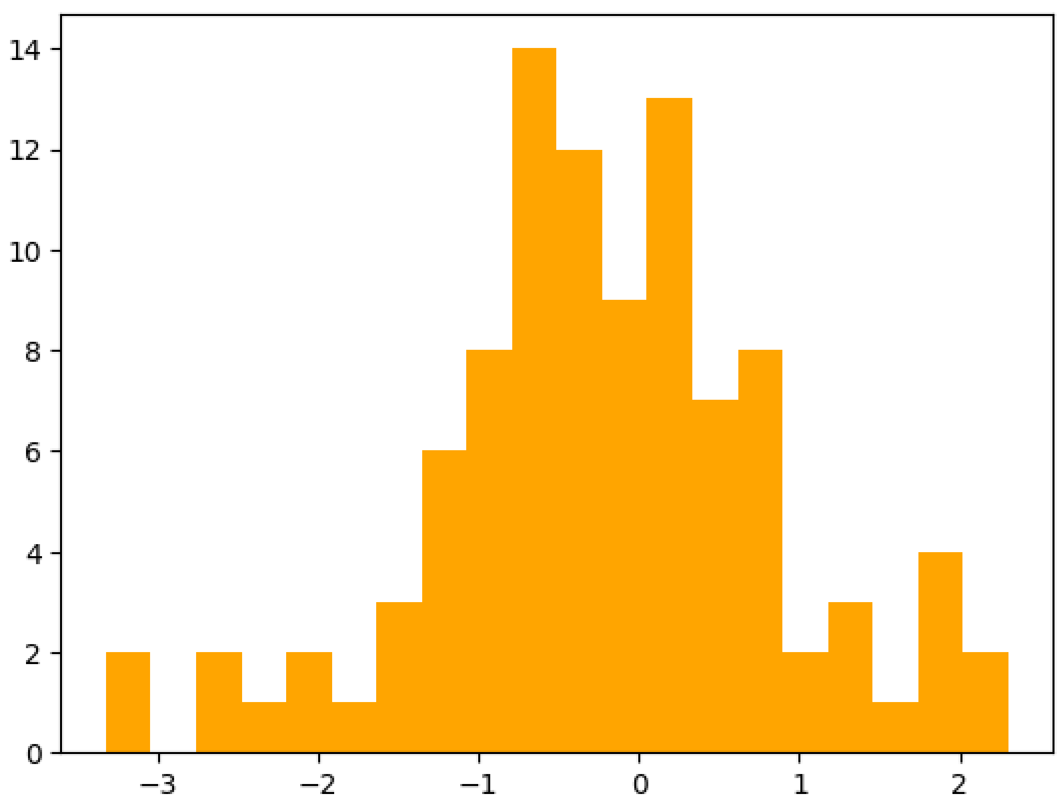
In [45]:
```
import numpy as np
```

In [46]:
```
data= np.random.randn(100)
```

In [47]:
```
data
```

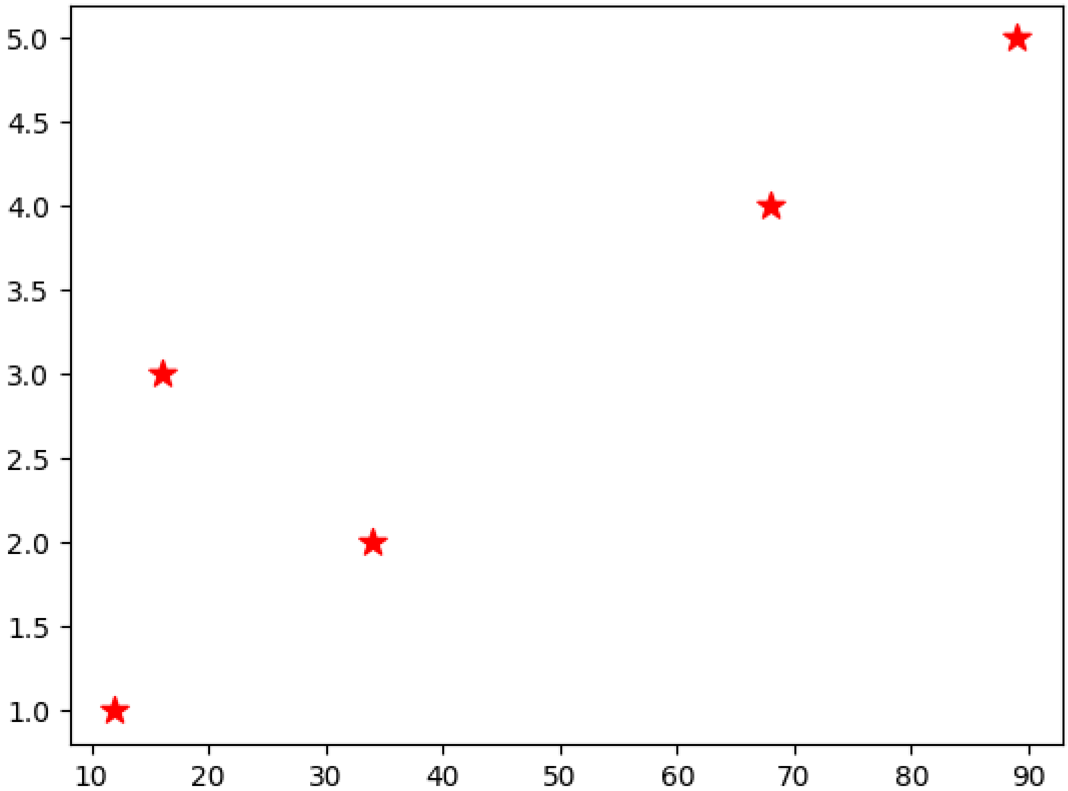
```
Out[47]: array([ 1.66409424e-01, -1.30557190e+00, -7.67557687e-01,  8.73355680e-01,
                -3.17936951e+00, -2.25830390e+00, -5.78812856e-01, -3.17046395e-01,
                -1.46348399e+00, -9.01587114e-01,  8.30595088e-02,  7.94031996e-01,
                -3.48938761e-01, -9.02107318e-01, -8.31588016e-01,  1.74828350e+00,
                 1.77266292e+00, -6.20175934e-01,  1.07379329e-01, -7.44607838e-01,
                 1.32783747e-01, -6.14609316e-01, -2.53051573e-01,  1.70375149e+00,
                -1.60671039e+00, -1.16232145e+00, -6.09666413e-01, -9.89915921e-02,
                 2.15433096e-02,  5.58678256e-01, -3.72870549e-01, -2.52856454e+00,
                -9.91226126e-02, -1.91907754e-01, -1.04702365e+00, -6.56262118e-03,
                -5.92774992e-01, -6.21927180e-01,  6.66047659e-01, -3.42726478e-01,
                -6.16623981e-02, -1.22307614e-01,  2.30254264e+00, -1.91886603e+00,
                -7.86395906e-01, -5.62839228e-01, -8.89549074e-01,  7.86642635e-01,
                 1.19585919e+00, -9.34265935e-01,  5.65995263e-01,  1.40858017e-01,
                 1.26739096e-01, -4.30725523e-01, -3.04126976e-01,  4.83202092e-01,
                 2.36063433e-01,  5.24605175e-01, -2.02531888e+00,  2.01679999e+00,
                -5.94159939e-01,  9.05375284e-01, -6.15817078e-02,  1.86293939e+00,
                -5.38645795e-01,  3.30121287e-01,  1.33123971e-01, -6.84786441e-01,
                 1.19092432e+00, -4.68798165e-01, -3.32372939e+00,  5.62133151e-01,
                -1.21353423e+00, -1.17220915e+00, -7.28369243e-01, -1.38389186e+00,
                 2.15391432e-01,  4.81478430e-01,  3.09097361e-01, -5.08765022e-01,
                -1.66000180e-03, -1.34049288e+00, -9.15687345e-01,  6.76056361e-01,
                -4.17238669e-01,  2.25770604e+00,  1.18653900e+00,  8.34812057e-01,
                 1.16699983e+00, -4.92600736e-01,  6.89015810e-01, -1.26314509e+00,
                -1.03000431e+00, -1.88982727e+00, -2.56717066e+00,  3.68644076e-01,
                 2.33398230e-01,  6.82875904e-01,  2.05066907e-01, -4.89263229e-01])
```

```
In [53]: plt.hist(data, color= "orange",bins=20)
         plt.show()
```



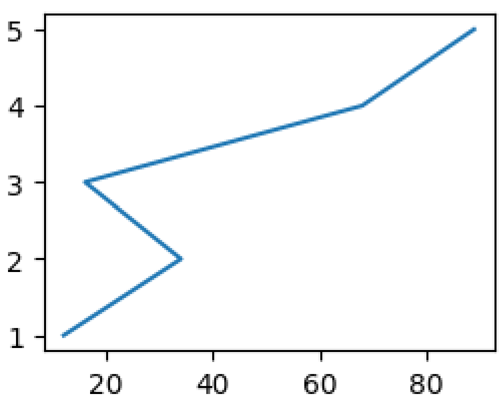
In [54]:
```
x=[12,34,16,68,89]
y=[1,2,3,4,5]
```

In [59]:
```
plt.scatter(x,y, color= "r", s=100, marker="*")
plt.savefig("scatter.png")
```



In [65]:
```
plt.subplot(2,2,1)
plt.plot(x,y)
```

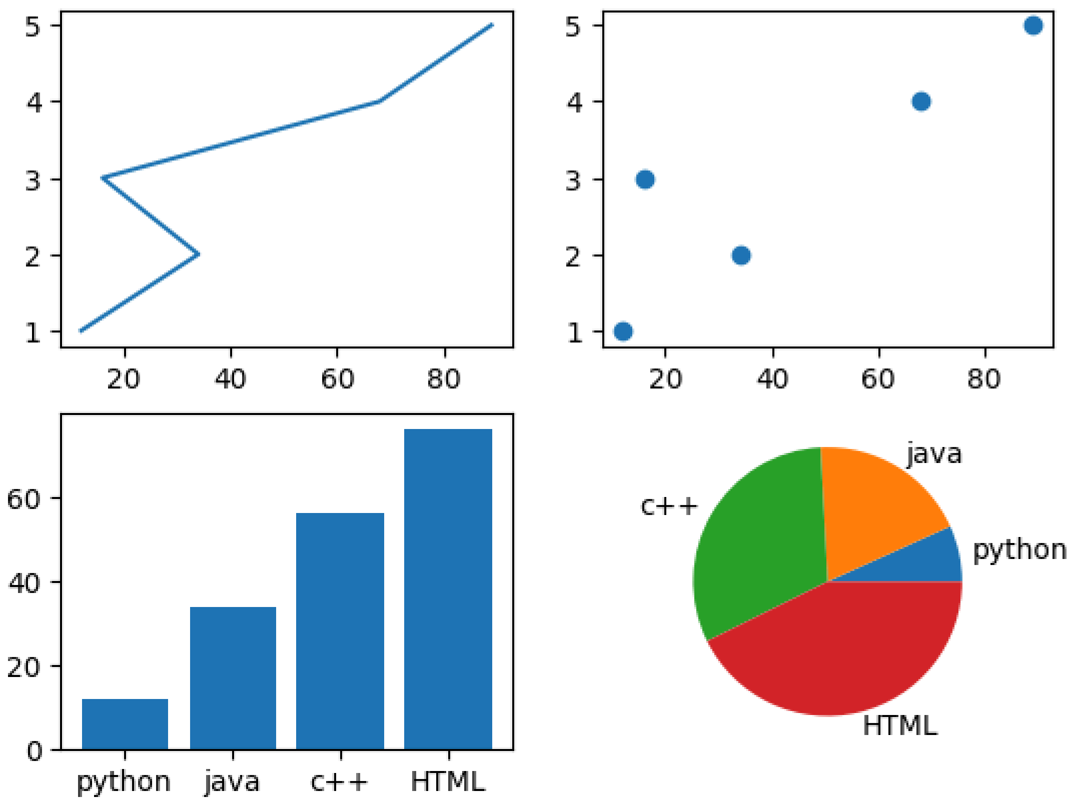
Out[65]: [<matplotlib.lines.Line2D at 0x298a66e64d0>]



In [67]:
```
plt.subplot(2,2,1)
plt.plot(x,y)
plt.subplot(2,2,2)
plt.scatter(x, y)
```

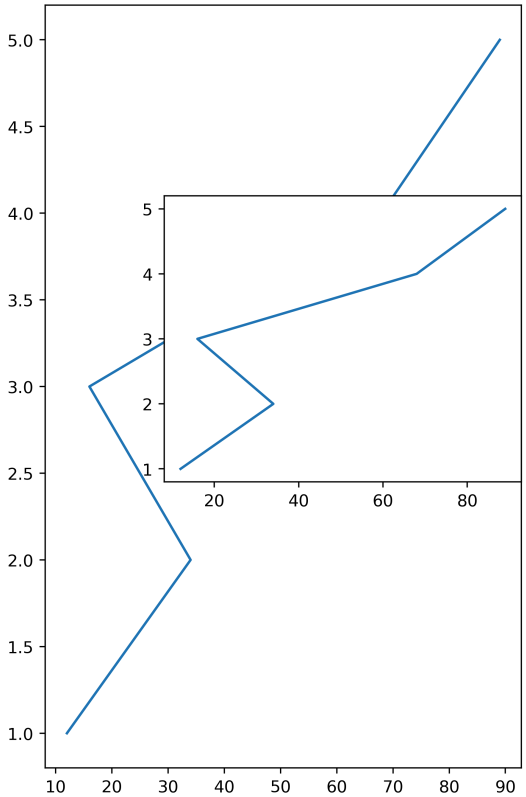
```python
plt.subplot(2,2,3)
plt.bar(x2,y2)
plt.subplot(2,2,4)
plt.pie(y2, labels=x2)
```

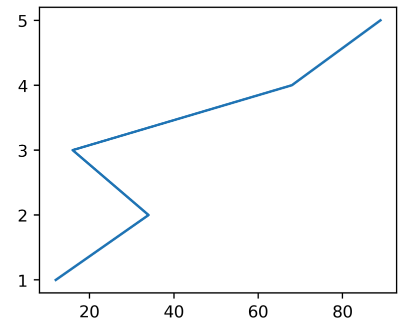
Out[67]:
```
([<matplotlib.patches.Wedge at 0x298a64e7bd0>,
  <matplotlib.patches.Wedge at 0x298a6b42950>,
  <matplotlib.patches.Wedge at 0x298a6b43890>,
  <matplotlib.patches.Wedge at 0x298a6b50a50>],
 [Text(1.0754211855014506, 0.23123424005681897, 'python'),
  Text(0.5722634598250345, 0.939422446266365, 'java'),
  Text(-0.9493758140193986, 0.5555947837723141, 'c++'),
  Text(0.25017774132941756, -1.071172767457851, 'HTML')])
```



In [83]:
```python
fig= plt.figure(figsize=(10,8), dpi=400)
axis1= fig.add_axes([0.1,0.1,0.4, 0.8])
axis1.plot(x,y)
axis2= fig.add_axes([0.8,0.4,0.3, 0.3])
axis2.plot(x,y)
axis3=  fig.add_axes([0.2,0.4,0.3, 0.3])
axis3.plot(x,y)
```

Out[83]:
```
[<matplotlib.lines.Line2D at 0x298a8e66790>]
```





In [69]:
```
fig
```

Out[69]: <Figure size 2000x800 with 0 Axes>

In [ ]:
```

```

Out[ ]: [<matplotlib.lines.Line2D at 0x298a6da66d0>]

In [84]:
```
import pandas as pd
```

In [85]:
```
df= pd.read_csv("data.csv")
```

In [86]:
```
df
```

Out[86]:

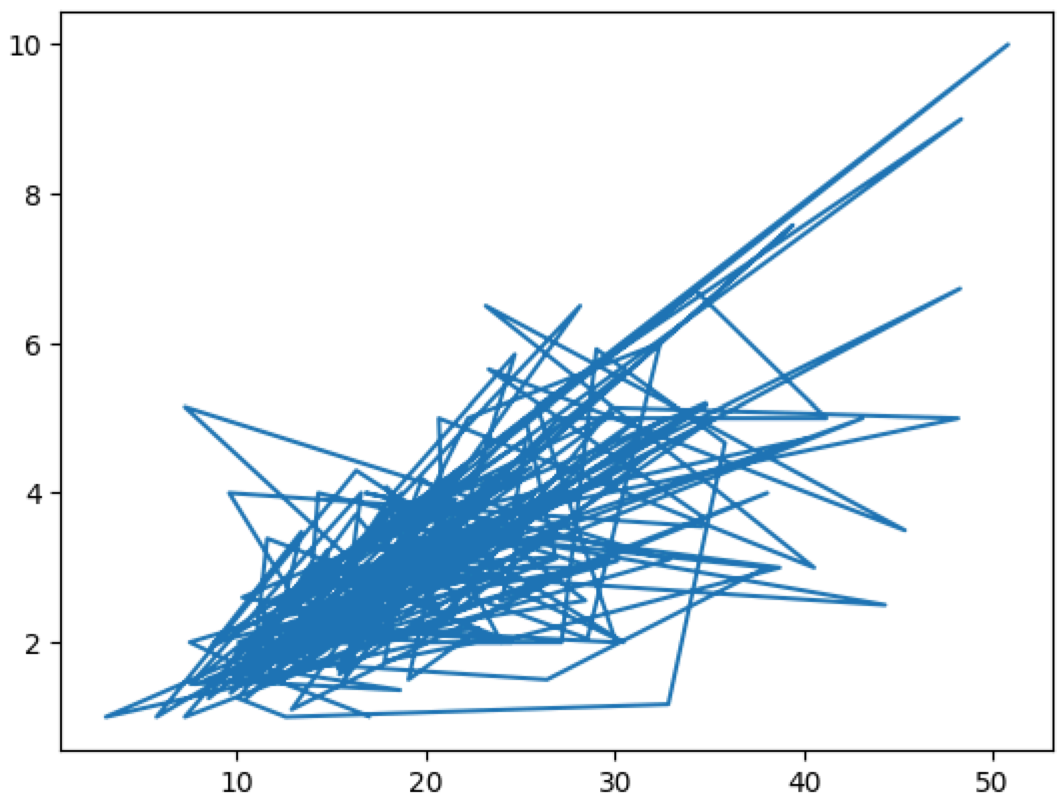
	total_bill	tip	gender	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

244 rows × 7 columns

In [87]:

```
plt.plot(df.total_bill, df.tip)
```

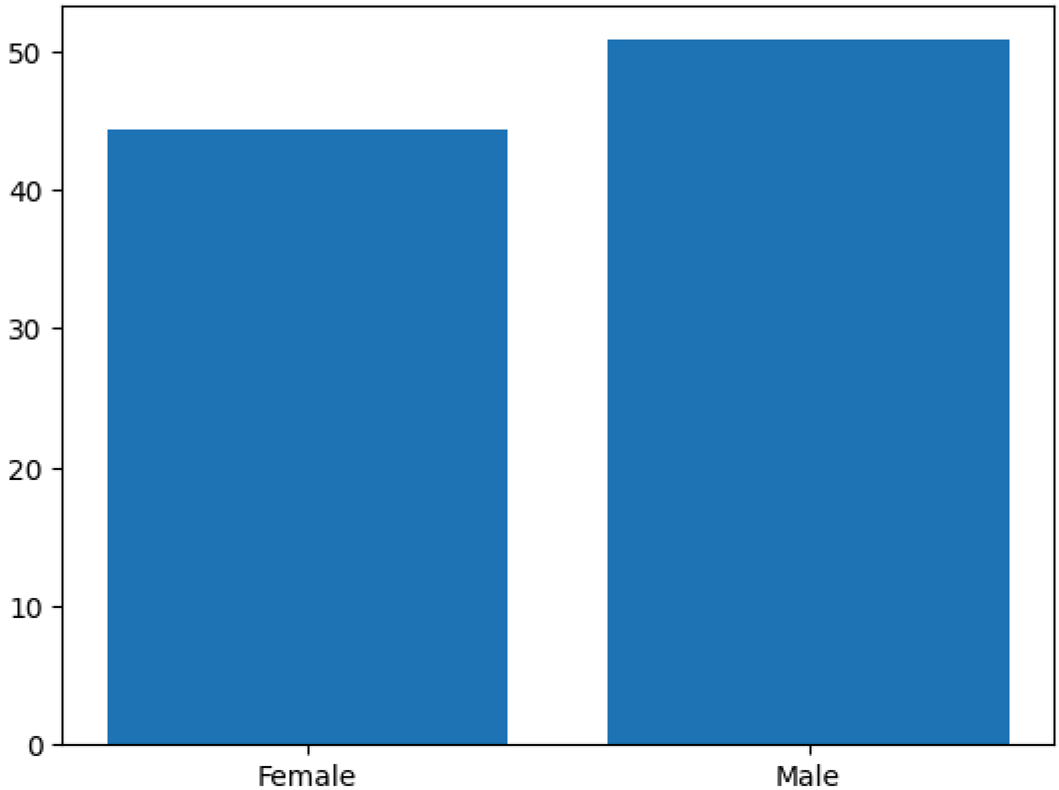
Out[87]: [<matplotlib.lines.Line2D at 0x298a8e9bc10>]



In [88]:

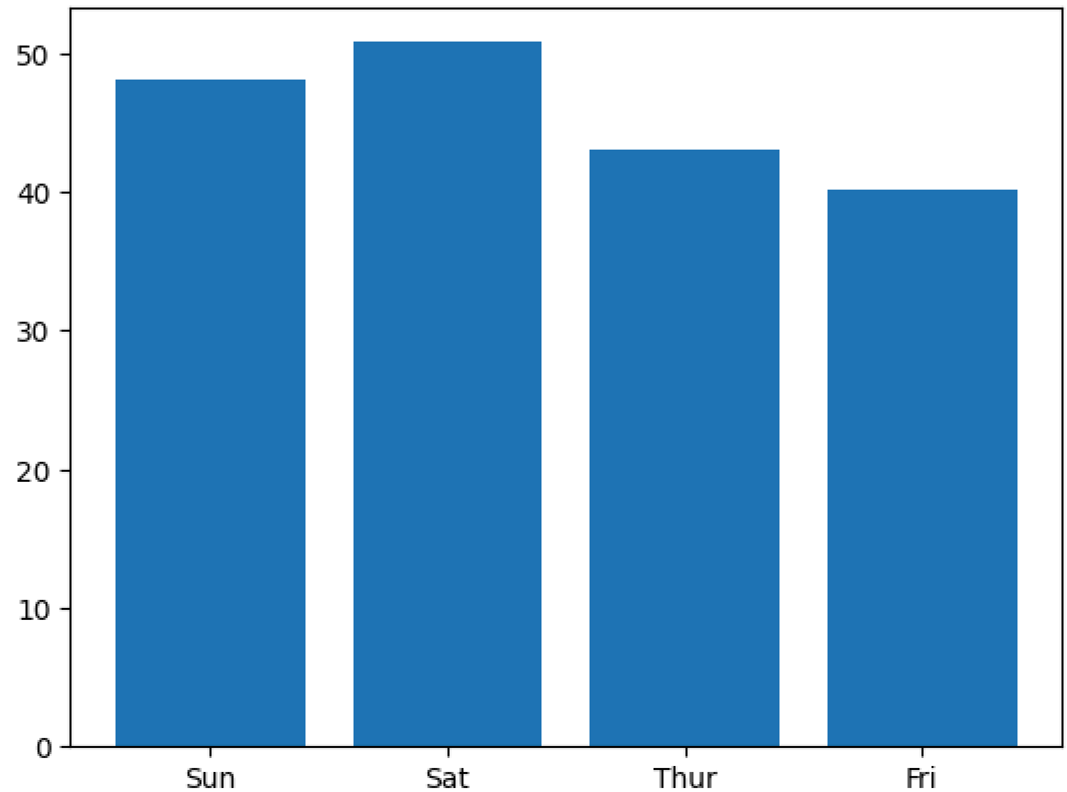
```
plt.bar(df.gender, df.total_bill)
```

Out[88]: <BarContainer object of 244 artists>



In [95]:
```
plt.bar( df.day,df.total_bill)
```

Out[95]: <BarContainer object of 244 artists>



In [90]:
```
df.day.unique()
```

Out[90]:
```
array(['Sun', 'Sat', 'Thur', 'Fri'], dtype=object)
```

In [105…
```
df2=df.groupby(["day"])["total_bill"].sum()
```

In [102…
```
df2
```

Out[102…
```
day
Fri      325.88
Sat     1778.40
Sun     1627.16
Thur    1096.33
Name: total_bill, dtype: float64
```

In [103…
```
df2[0]
```

```
C:\Users\rohan\AppData\Local\Temp\ipykernel_42300\19425605.py:1: FutureWarning:
Series.__getitem__ treating keys as positions is deprecated. In a future versio
n, integer keys will always be treated as labels (consistent with DataFrame beh
avior). To access a value by position, use `ser.iloc[pos]`
  df2[0]
```

Out[103…
```
np.float64(325.88)
```

In [106…
```
df2.index
```

Out[106…
```
Index(['Fri', 'Sat', 'Sun', 'Thur'], dtype='object', name='day')
```

In [107…

```
df2.values
```

Out[107…

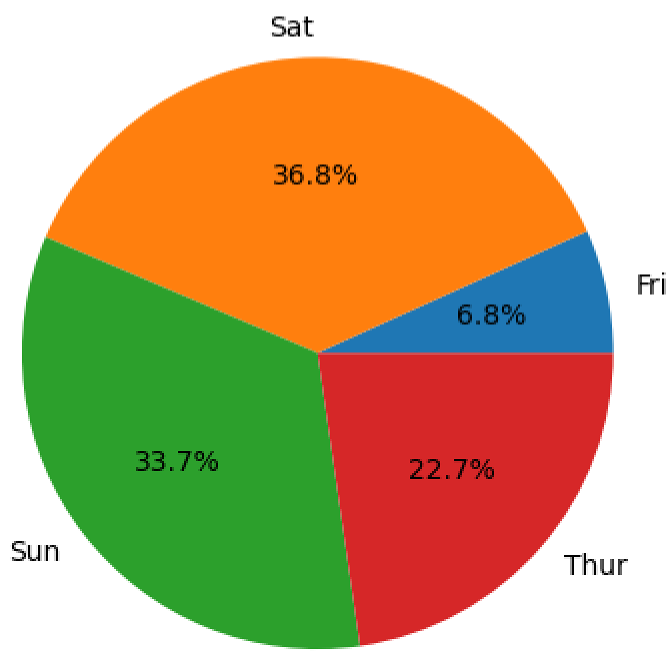
```
array([ 325.88, 1778.4 , 1627.16, 1096.33])
```

In [109…

```
plt.pie(df2.values, labels=df2.index, autopct="%1.1f%%")
```

Out[109…

```
([<matplotlib.patches.Wedge at 0x298b99f7590>,
  <matplotlib.patches.Wedge at 0x298b9a41750>,
  <matplotlib.patches.Wedge at 0x298b9a6c2d0>,
  <matplotlib.patches.Wedge at 0x298b9a6dcd0>],
 [Text(1.0753591008119494, 0.2315227943441335, 'Fri'),
  Text(-0.011649570046631373, 1.0999383107782619, 'Sat'),
  Text(-0.8717444419798231, -0.6708663263849862, 'Sun'),
  Text(0.8317422085432018, -0.7198645001162907, 'Thur')],
 [Text(0.5865595095337904, 0.12628516055134553, '6.8%'),
  Text(-0.006354310934352943, 0.5999663513335973, '36.8%'),
  Text(-0.4754969683526308, -0.3659270871190834, '33.7%'),
  Text(0.45367756829629186, -0.39265336369979487, '22.7%')])
```



In [ ]: